FACULTY OF COMPUTER APPLICATIONS

Marwadi University Marwadi Chandarana Group NAAC A+

Faculty of Computer Applications

BCA Sem-1 (AY-2026-27)

Lab Book

Sub Code: 05BC3103

(Programming Practices: 1)

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GitHub Link |https://github.com/Chauhancoding/Programing-Practic-1

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Program No	**Definition**
Unit 1	
1	Print " hello world".
2	Take 2 value from user and perform all arthemetic operation.
3	Print number from 1 to 20.
4	Input 3 value from user and find out its Average.
5	Calculate the square and cube of a number.
6	Find largest out of three number.
7	Find out Fibonacci series is odd and even.
8	Interchange the value of two number.
9	Find out number is odd or even.
10	Find out sum of digit of a given number.
Unit : 02	
1	Write a program to print " hello print"
2	Write a program that takes 2 value from user and preform all arthemetic operation.
3	Write a program that takes 3 value for principal amount, rate of interest and no of year and find out simple intersect.
4	Write a program that input from user and find out area of circle .
5	Write a program thar input two value from user and find out area of rectangle.
6	Write a program that input three value from user and find out its average .
7	Write a program that to calculate the square and cube of a number.
8	Write a program interchange the value of two number using third variable
9	Write a program interchange the value of two number without using third variable.

10	Write a program that input quantity, price, discount and display final amount.

.

Unit :01

01. Start
    .Printf"hello world" .Stop.

02. .Start
    .Read two number a ann b
    .Calculate sum = a+b
    .Calculate difference = a-b
    .Calculate product = a*b
    .If B is not equal to 0 , calculate quotient = a/b and remainder = a%b else display error . division by zero is not allowed
    .printf sum,difference,product,quotient and remainder. .stop

03. .start
    .open turbo c++
    .Create a new file
    .Set counter i=1
    .while i<=20, repeat step 4
    .print i
    .set i= i+1
    .stop

04 .Start

. input a number N

.find square N*N

.find cube N*N*N

. Display the square and cube and stop.

05 .start
    .Enter two number A and B
    .store A in temp
    . assign B to A
    . Assign tem to b
    . Display the new value of A and B
    . Stop

---

06 . .start
. Enter three number A,B,and C
. if A>=B and A>=C , set greatest = A
.otherwise if B>=C, ser greatest= B
. otherwise set greatest = C
. display the greatest number
. stop
.start

07. Enter the number of term N
. initialize first = 0 second = 1 amd count = 1
. repeat while count Fibonacci term
. calculate next = first+ second
.Update first = second and second = next
. increase count by 1
. stop
08. start
. input a number N
. check whether N is completely divisible by 2
. If yes , display “even”
. otherwise , display ” odd”
. stop
09. . start
. enter a position integer N
. initialize sum = 0
. repeat the following step until n become 0
. extract the last digital using digit = N%10
. Add digital to sum
. remove the last digit using N=N/10
. Display the final sum
. stop

10. ***Start.***

Read/Input the number N

Initialize a variable Sum = 0.

Loop while N is greater than 0:

Extract the last digit: Digit = N % 10 (Modulo operation gets the remainder) Update

the sum: Sum = Sum + Digit.

Remove the last digit: N = N / 10 (Integer division drops the remainder)

Print the value of Sum once the loop ends.

Stop.

Unit : 02

01.

```
#include <stdio.h>
#include<conio.h>

int main()
{
        printf("hello    world");
getch();
}
```

Output : hello

world

02.

```
#include <stdio.h>
#include<conio.h
> int main()
{
   float  num1, num2;
     clsrscr();       printf("Enter  first
number:    ");              scanf("%f",
&num1);       printf("Enter  second
number:    ");              scanf("%f",
&num2);             printf("Addition:
%.2f\n",      num1      +      num2);
```

```
printf("Subtraction:          %.2f\n",
num1                -                num2);
printf("Multiplication:       %.2f\n",
num1 * num2);

      if  (num2  !=  0)  {            printf("Division:
%.2f\n", num1 / num2);
   }
 Else
 {
       printf("Division: Cannot divide by zero\n");
   }

   Getch();
}
```

output :

Enter first number: 5

Enter second number: 4

Addition: 9.00

Multiplication: 20.00

Division: 1.25

.03

```
#include <stdio.h>
#include <conio.h>

int main()
 {
   float principal, rate, years, simple_interest;
clrscr();
   printf("Enter Principal Amount: ");
scanf("%f",                    &principal);
printf("Enter  Rate  of  Interest  :  ");
scanf("%f",  &rate);       printf("Enter
Number of Years: ");
   scanf("%f", &years);

   simple_interest = (principal * rate * years) / 100;

   printf("Simple Interest: %.2f\n", simple_interest);
   getch();
}
```

output :

Enter Principal Amount : 40

Enter Rate of Amount : 50

Enter Number of Years : 2

Simple Interest : 40.00 .04

```
#include <stdio.h>
#include<conio.h>

int main()
 {
   float radius, area;
clrscr();    const float PI =
3.14159;

   printf("Enter the radius of the circle: ");
scanf("%f", &radius);

   area = PI * radius * radius;
```

```
   printf("Area of the Circle: %.2f\n", area);
   getch ();
}
```

Output:
Enter the radius of the circle : 3
Area of the circle : 28.27

05

```
#include <stdio.h>

#include<conio.h> int

main()

{

   float length, width, area;

     clrscr();      printf("Enter the length of the

rectangle:   ");            scanf("%f",   &length);

printf("Enter  the  width  of  the  rectangle:  ");

scanf("%f", &width);


   area = length * width;


   printf("Area of the Rectangle: %.2f\n", area);

getch();

}
```

output:

Enter the length of the rectangle : 2

Enter the Width of the rectangle : 3

Area of the Rectangle : 6.00

06.

```
#include <stdio.h>
#include<conio.h>

int main()
 {
  Float n1, n2,n3, average;
    clrscr();
      printf("Enter the number n1: ");
scanf("%f “,&n1);          printf("Enter
the number n2: ");          scanf("%f
“,&n2);      printf("Enter the number
n3: ");     scanf("%f “,&n3);
   average = (val1 + val2 + val3) / 3;

   printf("Average: %.2f\n", average);
   getch();
}
```

Output:
Enter the number n1: 4
Enter tte number n2: 44
Enter the number n3 : 444
Average : 164.00

07.

```
#include <stdio.h>

#include<conio.h> int

main()

{    float a;    clrscr();

printf("Enter a number: ");

scanf("%f", &a);


   printf("Square of %f”, a * a);

printf("Cube of %f”, a * a * a);
```

```
   getch();
}
```

Output :

Enter a number: 3

Square of 9.000

Cube of 27.000

08.

```
#include <stdio.h>
#include<conio.h>

int main()
{    int a, b,
temp;
   clrsccr();    printf("Enter the
number: ");    scanf("%d",&a);
printf("\n Enter the number:");
   scanf(" %d",b );

   printf("\nBefore interchange: a = %d", a);
printf("\nBefore interchange: a = %d", b);

   temp = a;
a = b;
   b = temp;

   printf("After interchange: a = %d", a);
printf("After interchange: a = %d", b);
 getch();
}
```

Output :

Enter the value : 2

Enter the value : 3

Before interchange : a=2

Before interchange : b=3

After interchange : a=3

After interchange : b=2

09.

```
#include<stdio.h>
#include <conio.h> int
main()
{    int a, b;   printf("Enter  numbers a: ");
scanf("%d ", &a);  printf("\nEnter
numbers b: ");   scanf("%d “, &b);
printf("Before interchange: a = %d,  a);
printf("\nBefore interchange: b = %d,  b);

 a = a + b;     b = a - b;     a = a - b;
printf("After interchange: a = %d, a);
printf("After interchange: b = %d, b);

 getch();
}
```

Output:

Enter number a:2

Enter number b :3

Before interchange : a=2 Before

interchange : b= 3

After interchange : a= 3

After interchange : b=2

10.

```
#include <stdio.h>
#include<conio.h> int
main()
{
   float quantity, price, discount;
clrscr();
        float amount, discountAmount, finalAmount;
printf("Enter quantity: "); scanf("%f", &quantity);
printf("Enter price: "); scanf("%f", &price);
printf("Enter discount percentage: "); scanf("%f",
&discount); amount = quantity * price;
discountAmount = amount * discount / 100;
finalAmount = amount - discountAmount; printf("Total
Amount = %.2f\n", amount); printf("Discount Amount
= %.2f\n", discountAmount); printf("Final Amount =
%.2f\n", finalAmount); getch();
}
```

Output:

Enter quantity : 2

Enter price : 200

Enter discount percentage : 2

Total Amount = 400

Discount Amount = 8.00

Final amount 392.00